

Practice Problem 9.1

Given, $y = 30 \sin(4\pi t - 75^\circ)$

comparing with $y = A \sin(\omega t + \phi)$,

Amplitude, $A = 30$

Phase, $\phi = -75^\circ$

Angular frequency, $\omega = 4\pi = 12.57 \text{ rad/s}$

Period, $T = \frac{2\pi}{\omega} = \frac{2\pi}{4\pi} = \frac{1}{2} = 0.5 \text{ s}$

Frequency, $f = \frac{1}{T} = \frac{1}{0.5} = 2 \text{ Hz}$

Practice Problem 9.2

Given, $i_1 = -4 \sin(377t + 55^\circ) = 4 \cos(377t + 55^\circ + 90^\circ)$

$$\therefore i_1 = 4 \cos(377t + 145^\circ)$$

$$i_2 = 5 \cos(377t - 65^\circ)$$

$$\therefore \phi_1 = 145^\circ$$

$$\therefore \phi_2 = -65^\circ$$

$$\therefore \text{Phase angle between } i_1 \text{ and } i_2 = \phi_1 - \phi_2 = 145^\circ + 65^\circ = 210^\circ$$

Practice Problem 9.3

$$\begin{aligned} \text{a) } & [(5+j2)(-1+j4) - 5\angle 60^\circ]^* \\ & = [-5 + 20j - 2j + 8j^2 - 5(\cos 60^\circ + j\sin 60^\circ)]^* \\ & = [-5 + 18j - 8 - 5(\frac{1}{2} + j\frac{\sqrt{3}}{2})]^* \\ & = [-13 + 18j - \frac{5}{2} - j\frac{5\sqrt{3}}{2}]^* \\ & = [-15.5 + j13.67]^* \\ & = -15.5 - j13.67 \quad \underline{\underline{\text{Ans}}}\end{aligned}$$

$$\begin{aligned} \text{b) } & \frac{10 + j5 + 3\angle 40^\circ}{-3 + j4} + 10\angle 30^\circ + j5 \\ \therefore 3\angle 40^\circ & = 3(\cos 40^\circ + j\sin 40^\circ) = 3(0.77 + j0.64) \\ & = 2.3 + j1.93 \\ \therefore 10\angle 30^\circ & = 10(\cos 30^\circ + j\sin 30^\circ) = 10(\frac{\sqrt{3}}{2} + j\frac{1}{2}) \\ & = 8.66 + j5\end{aligned}$$

$$\begin{aligned} \therefore \frac{10 + j5 + 3\angle 40^\circ}{-3 + j4} + 10\angle 30^\circ + j5 & = \frac{10 + j5 + 2.3 + j1.93}{-3 + j4} + 8.66 + j5 + j5 \\ & = \frac{12.30 + j6.93}{-3 + j4} + 8.66 + j10 \\ & = \frac{(12.30 + j6.93)(-3 - j4)}{(-3)^2 + (4)^2} + 8.66 + j10\end{aligned}$$

$$= \frac{-36.9 - j49.2 - j20.79 + 27.72}{9 + j16} + 8.66 + j10$$

$$= \frac{-9.18 - j69.99}{25} + 8.66 + j10$$

$$= -\frac{9.18}{25} - \frac{j69.99}{25} + 8.66 + j10$$

$$= -0.3672 - j2.8 + 8.66 + j10$$

$$= 8.30 + j7.2$$

Ans

Practice Problem 9.4

a) $v = 7\cos(2t + 40^\circ) \text{ V}$

$$\therefore \bar{V} = 7 \angle 40^\circ$$

b) $i = -4\sin(10t + 10^\circ) \text{ A} = -4\sin(10t + 100^\circ - 90^\circ)$

$$= 4\cos(10t + 100^\circ)$$

$$\therefore \bar{I} = 4 \angle 100^\circ$$

Practice Problem 9.5:

a) Given, $\bar{V} = -25 \angle 40^\circ \text{ V} = 25 \angle 40^\circ + 180^\circ \text{ V} = 25 \angle 220^\circ \text{ V}$

$$\therefore v(t) = X_m \cos(\omega t + \phi)$$

Here, $X_m = 25$ and $\phi = 220^\circ$

$$\therefore v(t) = 25 \cos(\omega t + 220^\circ) \text{ V}$$

b) Given, $\bar{I} = j(12 - j5) \text{ A}$
 $= j12 + 5$

$$\therefore |\bar{I}| = \sqrt{5^2 + 12^2} = 13$$

$$\therefore \theta = \tan^{-1} \frac{12}{5} = 67.4^\circ$$

$$\therefore \bar{I} = 13 \angle 67.4^\circ$$

$$\therefore i(t) = 13 \cos(\omega t + 67.4^\circ) \text{ A}$$

Practice Problem 9.6:

$$\text{Given, } v_1 = -10 \sin(\omega t - 30^\circ) \text{ V}$$

$$v_2 = 20 \cos(\omega t + 45^\circ) \text{ V}$$

$$\therefore v_1 = -10 \sin(\omega t - 30^\circ)$$

$$~~= -10~~ \sin(\omega t + 60^\circ - 90^\circ)$$

$$= 10 \cos(\omega t + 60^\circ)$$

$$\therefore \bar{v}_1 = 10 \angle 60^\circ = 10 (\cos 60^\circ + j \sin 60^\circ) = 5 + j 8.66$$

$$\therefore \bar{v}_2 = 20 \angle 45^\circ = 20 (\cos 45^\circ + j \sin 45^\circ) = 14.14 + j 14.14$$

$$\therefore \bar{v} = v_1 + v_2 = 5 + j 8.66 + 14.14 + j 14.14$$
$$= 19.14 + j 22.8$$

$$\therefore |\bar{v}| = \sqrt{(19.14)^2 + (22.8)^2} = 29.8$$

$$\therefore \theta = \tan^{-1} \left(\frac{22.8}{19.14} \right) \approx 50^\circ = 49.98^\circ$$

$$\therefore v(t) = 29.8 \cos(\omega t + 50^\circ)$$
$$= 29.8 \cos(\omega t + 49.98^\circ)$$